Matgeo Presentation - Problem 4.13.58

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October 11, 2025

Problem Statement

Show that all chords of the curve $3x^2 - y^2 - 2x + 4y = 0$, which subtend a right angle at the origin, pass through a fixed point. Find the coordinates of the point. (1991)

Solution: Matrix Representation

The equation of the given curve is $3x^2 - y^2 - 2x + 4y = 0$
Let $\mathbf{x} = \begin{pmatrix} x \\ y \end{pmatrix}$.
The equation of the curve can be written in the matrix form
$\mathbf{x}^\top V\mathbf{x} + 2\mathbf{u}^\top \mathbf{x} + f = 0$ as

$$\mathbf{x}^\top \begin{pmatrix} 3 & 0 \\ 0 & -1 \end{pmatrix} \mathbf{x} + 2 \begin{pmatrix} -1 \\ 2 \end{pmatrix}^\top \mathbf{x} + 0 = 0. \tag{0.1}$$

Solution: Matrix Representation

Here, the matrix of the quadratic part is

$$V = \begin{pmatrix} 3 & 0 \\ 0 & -1 \end{pmatrix} \tag{0.2}$$

the vector for the linear part is

$$\mathbf{u} = \begin{pmatrix} -1 \\ 2 \end{pmatrix} \tag{0.3}$$

and the constant term is $f = 0$.

Solution: Homogenization

Let the equation of a chord PQ of the curve be $lx + my = 1$.
In vector form, this is:

$$\mathbf{n}^\top \mathbf{x} = 1, \quad \text{where } \mathbf{n} = \begin{pmatrix} l \\ m \end{pmatrix} \tag{0.4}$$

The combined equation of the pair of lines joining the origin to the points of intersection, P and Q, is obtained by homogenizing equation (0.1) using equation (0.4).

$$\mathbf{x}^\top V \mathbf{x} + 2(\mathbf{u}^\top \mathbf{x})(1) = 0 \tag{0.5}$$

Solution: Homogenization

Substituting $\mathbf{n}^\top \mathbf{x}$ for 1:

$$\mathbf{x}^\top V \mathbf{x} + 2(\mathbf{u}^\top \mathbf{x})(\mathbf{n}^\top \mathbf{x}) = 0 \tag{0.6}$$

This can be written as a single quadratic form:

$$\mathbf{x}^\top V \mathbf{x} + \mathbf{x}^\top(\mathbf{u}\mathbf{n}^\top + \mathbf{n}\mathbf{u}^\top)\mathbf{x} = 0 \tag{0.7}$$

$$\mathbf{x}^\top(V + \mathbf{u}\mathbf{n}^\top + \mathbf{n}\mathbf{u}^\top)\mathbf{x} = 0 \tag{0.8}$$

Solution: Condition for Perpendicularity

This is the equation of the pair of lines OP and OQ. Let

$$W = V + \mathbf{u}\mathbf{n}^\top + \mathbf{n}\mathbf{u}^\top. \tag{0.9}$$

The lines represented by the homogeneous quadratic equation $\mathbf{x}^\top W\mathbf{x} = 0$ are perpendicular if and only if the trace of the matrix W is zero.

Solution: Condition for Perpendicularity

The condition is $\text{trace}(W) = 0$. Using the property

$$\text{trace}(A + B) = \text{trace}(A) + \text{trace}(B) \tag{0.11}$$

$$\text{trace}(\mathbf{a}\mathbf{b}^\top) = \mathbf{b}^\top \mathbf{a} \tag{0.12}$$

We have:

$$\text{trace}(W) = \text{trace}(V) + \text{trace}(\mathbf{u}\mathbf{n}^\top) + \text{trace}(\mathbf{n}\mathbf{u}^\top) \tag{0.13}$$

$$\text{trace}(W) = \text{trace}(V) + \mathbf{n}^\top \mathbf{u} + \mathbf{u}^\top \mathbf{n} = \text{trace}(V) + 2\mathbf{u}^\top \mathbf{n} \tag{0.14}$$

Solution: Condition for Perpendicularity

The condition for the chord to subtend a right angle at the origin is

$$\text{trace}(V) + 2\mathbf{u}^\top \mathbf{n} = 0 \tag{0.17}$$

From the problem statement:

$$V = \begin{pmatrix} 3 & 0 \\ 0 & -1 \end{pmatrix} \tag{0.18}$$

$$\implies \text{trace}(V) = 3 + (-1) = 2 \tag{0.19}$$

Solution: Deriving the Condition

We also have:

$$\mathbf{u} = \begin{pmatrix} -1 \\ 2 \end{pmatrix} \tag{0.20}$$

$$\mathbf{n} = \begin{pmatrix} l \\ m \end{pmatrix} \tag{0.21}$$

Substituting these values into equation (0.17)

$$2 + 2 \begin{pmatrix} -1 & 2 \end{pmatrix} \begin{pmatrix} l \\ m \end{pmatrix} = 0 \tag{0.22}$$

Solution: Deriving the Condition

Simplifying the expression:

$$2 + 2(-l + 2m) = 0 \tag{0.1}$$

$$1 - l + 2m = 0 \tag{0.2}$$

$$l - 2m = 1 \tag{0.23}$$

This is the condition that the coefficients of the chord $lx + my = 1$ must satisfy.

Solution: Finding the Fixed Point

We want to show that this line passes through a fixed point (x_0, y_0). If it does, then the coordinates of this point must satisfy the equation of the line for any l and m that satisfy condition (0.23).

$$lx_0 + my_0 = 1 \tag{0.24}$$

From condition (0.23), we express l in terms of m: $l = 1 + 2m$. Substituting this into equation (0.24):

$$(1 + 2m)x_0 + my_0 = 1 \tag{0.25}$$

Solution: Finding the Fixed Point

Expanding the equation:

$$x_0 + 2mx_0 + my_0 = 1 \tag{0.26}$$

$$(x_0 - 1) + m(2x_0 + y_0) = 0 \tag{0.27}$$

Since this equation must hold for all possible values of m, the coefficients must be individually zero.

$$x_0 - 1 = 0 \tag{0.28}$$

$$\implies x_0 = 1 \tag{0.29}$$

Solution: Finding the Fixed Point

And for the second coefficient:

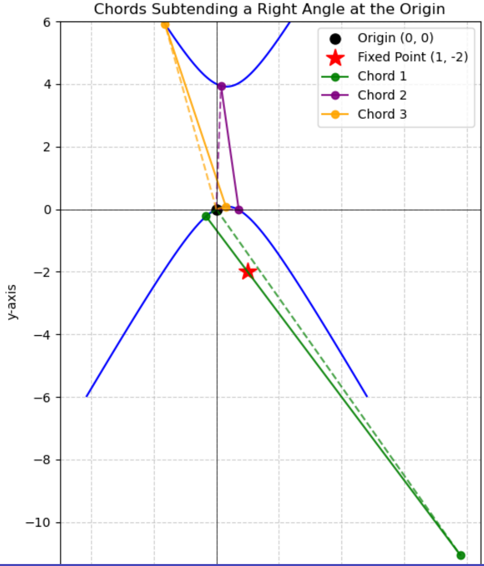
$$2x_0 + y_0 = 0 \tag{0.30}$$

$$\implies 2(1) + y_0 = 0 \tag{0.3}$$

$$\implies y_0 = -2 \tag{0.31}$$

Therefore, all such chords pass through the fixed point with coordinates $\begin{pmatrix} 1 \\ -2 \end{pmatrix}$.

Plot


Chords Subtending a Right Angle at the Origin
y-axis
6
4
2
0
−2
−4
−6
−8
−10
Origin (0, 0)
Fixed Point (1, -2)
Chord 1
Chord 2
Chord 3